Vibration Compensation for a Vehicle-Mounted Atom Gravimeter

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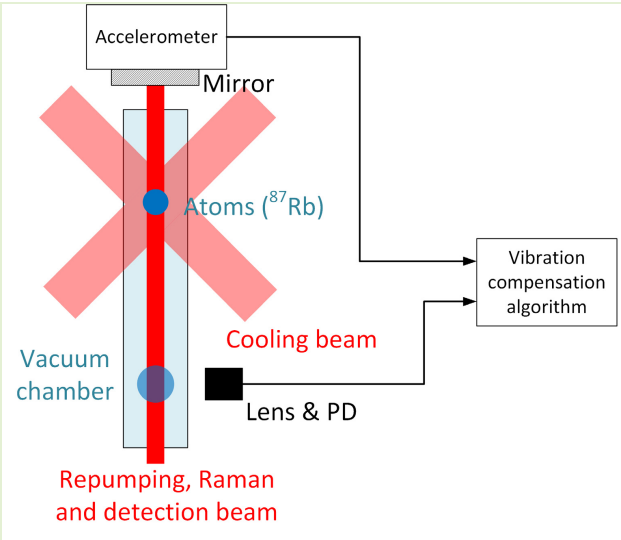
***Abstract*—The performance of absolute atom gravimeters used on moving platforms, such as vehicles, ships and aircrafts, is strongly affected by the vibration noise. To suppress its influence, we introduce a vibration compensation method, in which a classical accelerometer is used to measure the vibration noise. The measurement results obtained by the accelerometer show the vibration noise in the vehicle can be 2 orders of magnitude greater than that in the lab during daytime, and can induce an interferometric phase fluctuation with a standard deviation of 16.70π. With the compensation method, our vehicle-mounted atom gravimeter can work normally under harsh conditions. Comparing the Allan standard deviations before and after the vibration noise correction, we find a suppression factor of 22.74 can be achieved in static condition with a half interrogation time of T = 20 ms, resulting in a sensitivity of 1.35 mGal/Hz$^{1/2}$, and a standard deviation of 0.5 mGal with an average time of 10 s. We also demonstrate the first test of an atom gravimeter in a moving vehicle, in which a suppression factor of 50.85 and a sensitivity of 60.88 mGal/Hz$^{1/2}$ were realized with T = 5 ms.**



***Index Terms*—Atom gravimeter, vibration compensation, vehicle-mounted.**

I. Introduction

GRAVIMETRY plays an important role in metrology, navigation, geodesy, geophysics [1], [2], etc. After M. Kasevich and S. Chu demonstrated the measurement of gravity with their atom interferometer using three Raman pulses in 1991 [3], [4], many institutes began to develop atom gravimeters that can be used outside laboratories with similar configurations. And their performance now rivals the performance of the best gravimeters of any kind [5]. In recent years, AOSense, Inc. [6] and Muquans [7] have made commercial atom gravimeters. With gravimeters developed by Muquans fixed at the measurement location, the precision of the gravity measurement can reach μGal level [7]. Besides, atom gravimeters can also be used in absolute airborne and marine gravimetry, with mGal and submGal precisions respectively [8], [9]. It needs to be noted that, unlike relative gravimeters, i.e., gravimeters based on micro-electro mechanical systems (MEMS) [10], [11], superconducting [12], and springs [13], atom gravimeters can measure absolute gravity directly, which makes gravity surveys more efficient. And this can lead to wider applications. For example, such kind of gravimeters can be used as reference gravimeters to calibrate relative gravimeters [11], or as a golden standard for inertial measurement units (IMU) that can be applied in many fields, such as navigations [14] and diagnostics [15].

However, atom gravimeters are very sensitive to vibrations. The interference signal of an atom gravimeter is proportional to the cosine of the acceleration with a period equal to $\lambda/2T^2$ [9], which is small compared to the variations of acceleration induced by vibrations when large interrogation time $2T$ is used to increase the sensitivity. This would lead to an ambiguity to determine the acceleration from the inter-

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ference signal [9]. Therefore, precise measurement of gravity cannot be performed without any suppression of vibration noise [16]. The suppression can be achieved with vibration-isolation systems, which can be passive [17] or active [18]. But these systems are bulky, and unsuitable for operation in noisy or mobile environments because of their low frequency resonances [19]. Another approach is to correct their impact on the gravity measurement by measuring them [20] with an accelerometer [21], [22], seismometer [16], [20], [22] or laser interferometer [23]. The vibration compensation technique has already been used by Muquans in their gravimeters [21]. As a result, their gravimeters are robust against seismic noise without the need of any isolation devices [21]. However, there still lacks detailed discussion about the application of this method in mobile environments. Furthermore, to our knowledge, though atom gravimeters have been tested in a stopped truck [1], [24], on a moving ship [9] and on an airplane in flight [8], there is no report so far about testing an atom gravimeter on a moving vehicle.

As a complement, here a vibration compensation method is summarized and demonstrated with a vehicle-mounted atom gravimeter, in which the acceleration of the mirror is measured with an accelerometer. Our experiments were conducted under both static and dynamic conditions. The vibration noise levels and the suppression factors of the vibration compensation in these conditions are first compared in detail, which provides supplementary information for developing atom gravimeters used in noisy or mobile environments. Besides, the results are also useful for developing gyroscopes [25] and accelerometers [26], [27] based on atom interferometry that are designed to be used in mobile environments, in which the vibration compensation method may also be applied.

II. Experimental Setup and Methods

A. Experimental Setup

The atom gravimeter is composed of a sensor head, a gyro-stabilized platform, a laser system and control electronics, which are shown in Fig. 1(a). The platform can maintain the sensor head aligned with the gravitational acceleration with a precision of 0.02°, using an integrated IMU. Fig. 1(b) shows the modules installed on the vehicle. The platform and the sensor head are mounted around the center of the vehicle, and a UPS (uninterruptible power supply) is mounted in the back to power the atom gravimeter. The vehicle has a diesel engine that can drive the integrated generator to charge the UPS. A GPS (Global Positioning System) antenna is mounted on the top of the vehicle, which is connected to the GPS receiver to calibrate the integrated IMU of the gyro-stabilized platform.

The schematic diagram of the sensor head is illustrated in Fig. 2. A standard three-dimensional magneto-optical trap (3D MOT) cools and collects $\sim 10^8$ ^{87}Rb atoms released from a dispenser in 130 ms. The atoms are then further cooled down to 5 μK in an optical molasses. Afterwards, the cold atoms are released in free fall. The bias coils create a vertical magnetic bias field to define a quantization axis. The atoms are then transferred to $|F = 1, m_F = 0\rangle$ by a 110 μs long microwave π pulse for state preparation. The atoms remaining in $|F = 2\rangle$ are blown away by a laser pulse afterwards. After

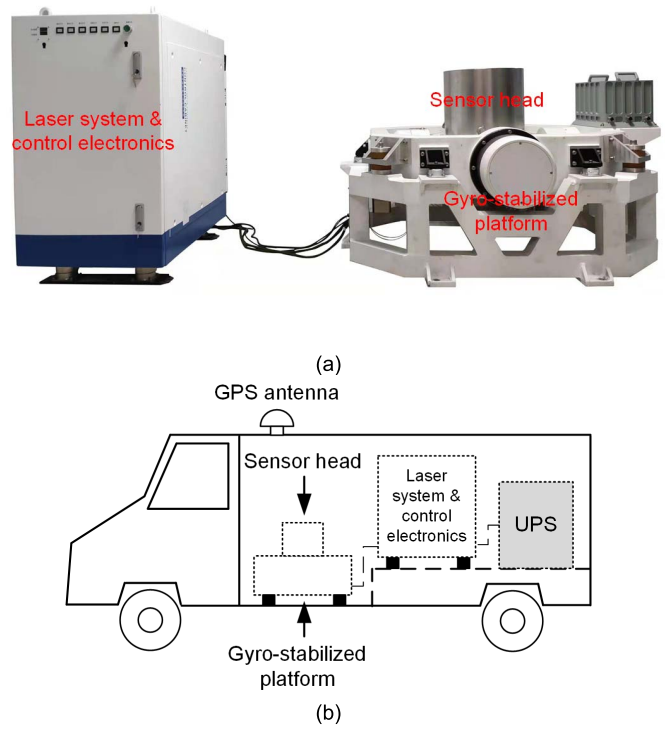

Fig. 1. (a) The photograph of the atom gravimeter. (b) Schematics of the modules installed on the vehicle.

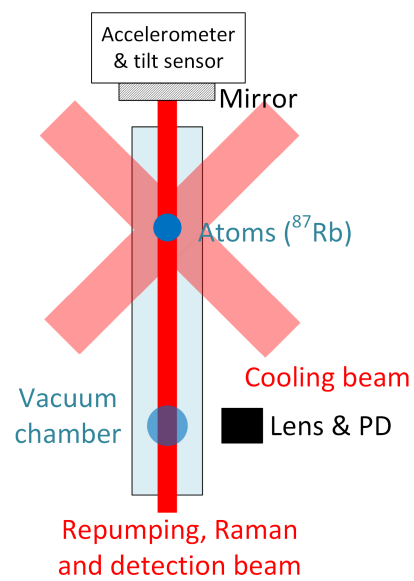

Fig. 2. The schematic diagram of the sensor head.

several milliseconds, the $\pi/2 - \pi - \pi/2$ sequence of Raman laser pulses are applied to split, reflect and recombine the matter waves. At the bottom of the vacuum chamber, the interference signal of the atom interferometer is obtained by measuring the population of atoms in $|F = 2\rangle$ and $|F = 1\rangle$ with a fluorescence method. The photodiode (PD), together with a lens beside the chamber, is used for the fluorescence detection. The mirror on the top is attached to a tilt sensor and an accelerometer. The tilt sensor is used to monitor

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the tilt of the Raman laser beam in static condition. And the accelerometer is used for vibration compensation. The sensor head is protected from external magnetic fields by two layers of mu-metal shields. And the outer layer can be seen in Fig. 1(a).

After the Raman pulses, the proportion P of the atoms in $|F=2\rangle$ is given by [9]

$$P = P_m - \frac{C}{2}\cos\Phi \tag{1}$$

where C is the contrast, P_m is the offset and Φ is the interferometric phase shift. The phase shift Φ can be written as

$$\Phi = (k_{eff}g - 2\pi\alpha)T^2 + \Phi_{vib} \tag{2}$$

in which $k_{eff} = 4\pi/\lambda$ is the effective wave vector of the Raman beam, g is the gravitational acceleration, α is the frequency chirp applied to the Raman lasers to compensate the Doppler shift induced by gravity, Φ_{vib} is the phase shift induced by vibrations, and T is the time interval between the 3 Raman pulses, namely the half interrogation time. With (1) and (2), we are able to deduce g from P and Φ_{vib}.

B. Vibration Compensation Method

As shown in Fig. 2, the Raman beam enters the vacuum chamber from the bottom. The vertical laser beam and its retro-reflection on the mirror mounted on top of the chamber form the two counter-propagating Raman beams. The phase difference between the two counter-propagating Raman beams is therefore defined by the position of the retro-reflection mirror [28]. A vertical displacement of this mirror δz induces a Raman phase shift of $k_{eff}\delta z$ [22]. Therefore, the phase shift induced by vibrations of the mirror can be calculated with the following equation [16], [22]:

$$\Phi_{vib} = k_{eff}\int_{-\infty}^{\infty} g_s(t)v_z(t)dt \tag{3}$$

where $v_z(t) = d\delta z/dt$ is the vertical velocity of the mirror, and $g_s(t)$ is the sensitivity function, given by [29]

$$g_s(t) = \begin{cases} 0 & t < -2\tau - T \\ \sin(\Omega_{eff}(t+T)) & -2\tau - T \le t < -\tau - T \\ -1 & -\tau - T \le t < -\tau \\ \sin(\Omega_{eff}t) & -\tau \le t < \tau \\ 1 & \tau \le t < \tau + T \\ \sin(\Omega_{eff}(t-T)) & \tau + T \le t < 2\tau + T \\ 0 & t \ge 2\tau + T \end{cases} \tag{4}$$

where $\Omega_{eff} = \pi/(2\tau)$ is the effective Rabi frequency.

In our setup, we use an accelerometer (Nanometrics Titan Accelerometer) attached to the mirror to measure its vibrations. Since it measures the acceleration instead of the velocity, we change (3) into the following form [26]:

$$\Phi_{vib} = k_{eff}\int_{-\infty}^{\infty} g_a(t)a_z(t)dt \tag{5}$$

TABLE I
THE STANDARD DEVIATIONS OF VIBRATIONS IN DIFFERENT PLACES UNDER SEVERAL TYPICAL CONDITIONS

Places	Speed (km/h)	σ_x (m/s^2)	σ_y (m/s^2)	σ_z (m/s^2)
Ground in the lab (daytime)	/	0.0014	0.0011	0.0016
Base plate on the vehicle (engine off)	/	0.0084	0.0074	0.0063
Mirror in the sensor head (engine off)	/	0.0052	0.0054	0.0074
Base plate on the vehicle (engine on)	0	0.1381	0.0921	0.0449
Mirror in the sensor head (engine on)	0	0.0313	0.0257	0.0286
Mirror in the sensor head (engine on)	7	0.0394	0.0553	0.0888
Mirror in the sensor head (engine on)	40	0.1494	0.1018	0.1686

where $a_z(t)$ is the vertical acceleration of the mirror, and $g_a(t)$ is the acceleration sensitivity function. Comparing (3) and (5), we find $g_a(t)$ can be expressed as

$$g_a(t) = \int_{-\infty}^{t} g_s(t)dt = \begin{cases} 0 & t < -2\tau - T \\ \dfrac{\cos(\Omega_{eff}(t+T)) + 1}{\Omega_{eff}} & -2\tau - T \le t < -\tau - T \\ \dfrac{1}{\Omega_{eff}} + t + \tau + T & -\tau - T \le t < -\tau \\ \dfrac{1 + \cos(\Omega_{eff}t)}{\Omega_{eff}} + T & -\tau \le t < \tau \\ \dfrac{1}{\Omega_{eff}} + \tau + T - t & \tau \le t < \tau + T \\ \dfrac{\cos(\Omega_{eff}(t-T)) + 1}{\Omega_{eff}} & \tau + T \le t < 2\tau + T \\ 0 & t \ge 2\tau + T \end{cases} \tag{6}$$

which is a triangle-like function. With (5) and (6), we are able to apply a post-correction.

III. RESULTS

A. Vibration Noise

To have a better knowledge of the vibration noise on the vehicle, we measured the vibrations in different places under several typical conditions, and made a comparison of their standard deviations in Table I. The vibrations of the ground and the base plate were measured with a strong motion accelerograph (Nanometrics TitanSMA), in which the sample rate was 2 kHz. The base plate refers to the plate on the vehicle used to fix the gyro-stabilized platform. And the vibrations of the mirror were measured with the accelerometer inside the atom gravimeter, in which the sample rate was several tens of kilohertz. The z axes of the accelerometer and the strong motion accelerograph were aligned with the gravitational acceleration.

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There are four typical conditions for the vehicle-mounted atom gravimeter. In the first case, the engine of the vehicle is off. The gravimeter is powered up with the UPS. The vibrations are minimized, which makes it possible to measure gravity with the longest interrogation time and the highest precision. However, the standard deviations of the vibrations in this case are still several times larger than the ones measured in the lab. In the second case, the engine is on for generating electricity but the vehicle doesn't move. In this case the gravimeter can work for a much longer time. The platform together with the sensor head may significantly attenuate the vibrations induced by the engine. In the third case, the vehicle moves with a speed of 7 km/h. The gravimeter can be used to measure gravity continuously along the path. In the last case, the vehicle moves with a speed of 40 km/h, which is a typical speed for transportation. Obviously, the vibration noise on the vehicle can be nearly 1 or 2 orders of magnitude greater than that in the lab, which makes the vibration compensation challenging. And the vibration noise in the last case can be comparable with that in the aircraft used by R. Geiger *et al.* [26].

The influence of the vibration noise on the interferometric phase shift can be estimated from its power spectral density (PSD). According to [28], the rms standard deviation of the interferometric phase noise $\sigma_{\Phi_{vib}}^{rms}$ induced by vibrations can be evaluated by

$$(\sigma_{\Phi_{vib}}^{rms})^2 = \int_0^{\infty} |H(\omega)|^2 \frac{k_{eff}^2}{\omega^4} S_a(\omega) d\omega \tag{7}$$

where $S_a(\omega)$ is the PSD of the vertical vibrations of the mirror. The phase transfer function $H(\omega)$ is given by [22], [28]

$$H(\omega) = \frac{4i\omega\Omega_{eff}}{\omega^2 - \Omega_{eff}^2} \sin(\frac{\omega(T+2\tau)}{2}) \times (\cos(\frac{\omega(T+2\tau)}{2}) + \frac{\Omega_{eff}}{\omega} \sin(\frac{\omega T}{2})) \tag{8}$$

For the duration of our Raman pulses, the complex magnitude of the acceleration-transfer function $|H_a(2\pi f)| = k_{eff}|H(2\pi f)|/(2\pi f)^2$ versus the frequency f is displayed in Fig. 3. Obviously, the function acts as a low pass filter.

With the measured data, we calculated the square roots of the PSDs of the vertical vibrations of the mirror. Fig. 4 shows the results obtained with the LPSD (Logarithmic frequency axis Power Spectral Density) method [30], [31]. With these spectra and (7), the influence of the vibration noise on the interferometric phase shift under the above mentioned 4 typical conditions is estimated and shown in Table II.

The standard deviation of the phase noise for T = 5 ms induced by vibrations measured while the vehicle was moving with a speed of 7 km/h is over 10π, which is 2 times larger than the value for T = 20 ms in static condition while the engine was off, and more than an order of magnitude larger than the typical value for T = 60 ms in Muquans laboratory [21]. This makes the moving test challenging.

B. Vibration Noise Correction

In our experiments, the vibration noise was post corrected. The frequency chirp α applied to the Raman lasers was fixed. The half interrogation time T is set to be 5 ms, 10 ms, or 20 ms, depending on the environmental conditions. Considering the z axes of the accelerometer might be misaligned with the Raman beam, the output signals along all its three axes were recorded by the control electronics of the

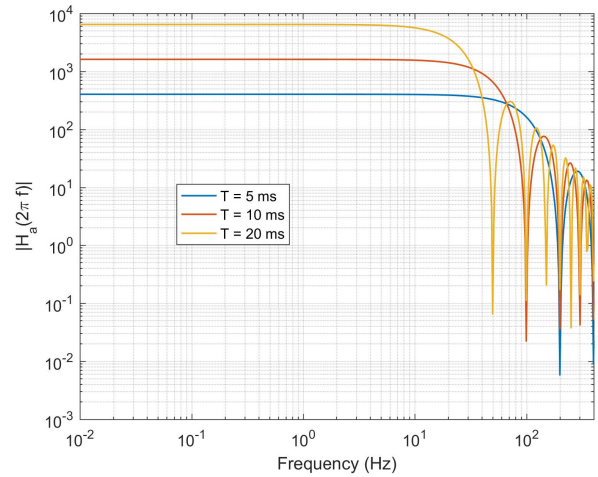

Fig. 3. The complex magnitude of the acceleration-transfer function $|H_a(2\pi f)|$ versus the frequency f.

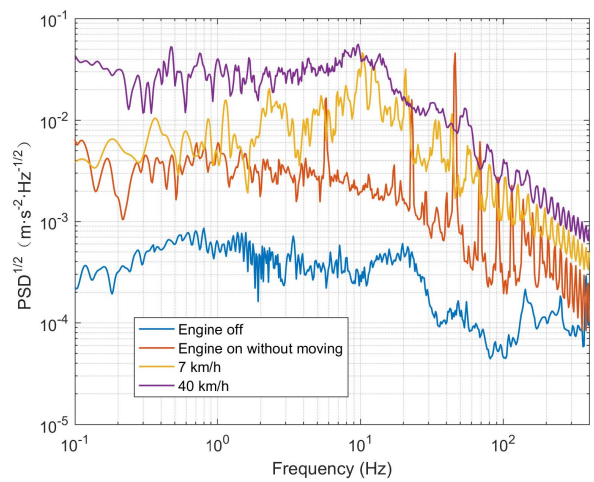

Fig. 4. Square roots of the PSDs of vertical vibrations of the mirror under the four typical conditions.

TABLE II
THE rms STANDARD DEVIATION OF THE INTERFEROMETRIC PHASE NOISE $\sigma_{\Phi_{vib}}^{rms}$ INDUCED BY VERTICAL VIBRATIONS OF THE MIRROR UNDER THE 4 TYPICAL CONDITIONS (ALL THE VALUES ARE IN RADIAN)

	T = 5 ms	T = 10 ms	T = 10 ms
Engine off	0.29π	1.08π	3.57π
Engine on without moving	6.09π	15.41π	29.40π
Engine on with a moving speed of 7 km/h	10.80π	40.10π	133.96π
Engine on with a moving speed of 40 km/h	20.51π	76.41π	265.32π

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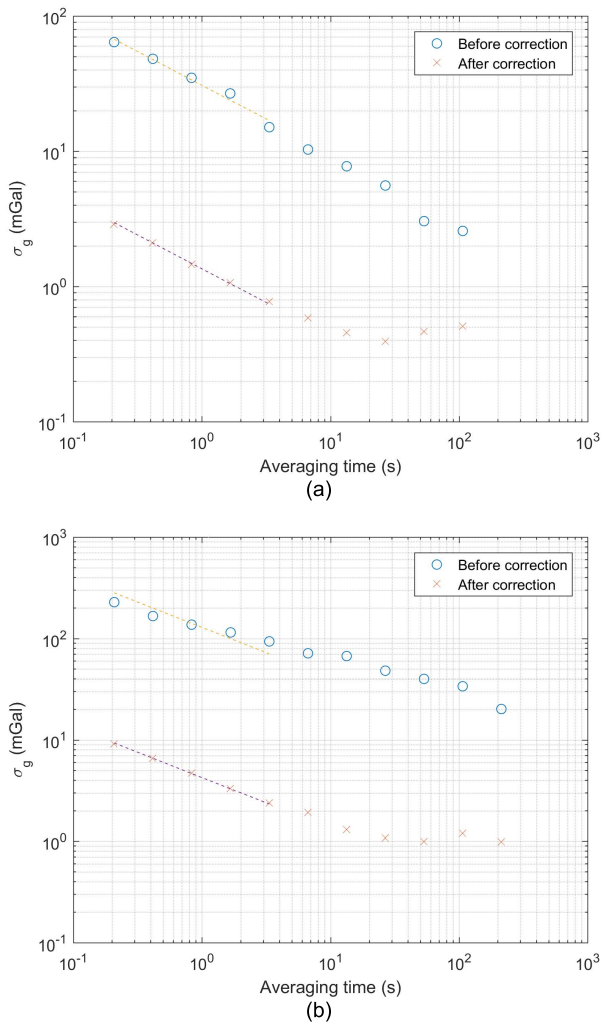

Fig. 5. The Allan standard deviation of the gravitational acceleration measured with the vehicle's engine off. (a) T = 20ms. (b) T = 10 ms.

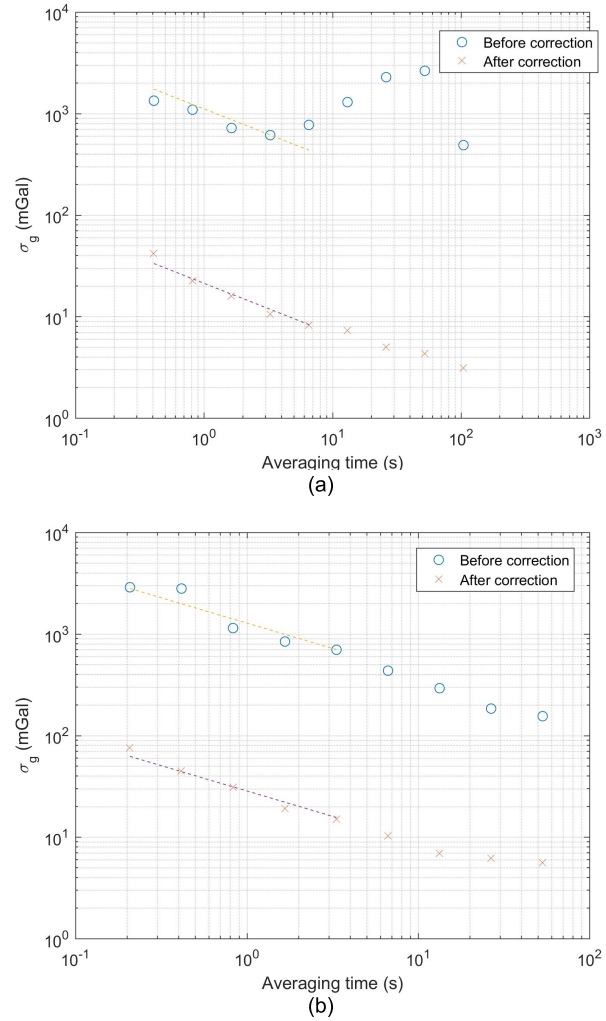

Fig. 6. The Allan standard deviation of the gravitational acceleration measured with the vehicle's engine on for generating electricity only. (a) T = 10ms. (b) T = 5 ms.

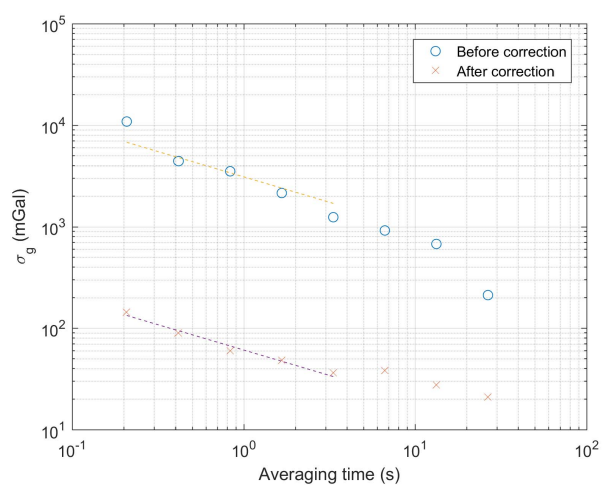

Fig. 7. The Allan standard deviation of the gravitational acceleration measured when the vehicle was moving with a speed of 7 km/h.

atom gravimeter with a sample rate of 62.5 kHz. The data acquisition is therefore synchronized with the time sequence of the interferometer. A high pass filter, whose cut-off frequency is 0.03 Hz, is used to eliminate the bias of the accelerometer output. After the experiments were finished, we are able to calculate the phase induced by vibration with $\Phi_{vib} = \sum \eta_j \phi_{vib,j}$, where $j = x, y, z$ and $\phi_{vib,j} = k_{eff} \int_{-\infty}^{\infty} g_a(t - \tau_j) a_j(t) dt$ is the phase shift calculated from the accelerometer data along axis j [20]. The contribution η_j and delay τ_j along axis j, together with the fringe offset P_m and contrast C, can be determined by fringe fitting, in which the proportion P of the atoms in $|F = 2\rangle$ is the observed output and the accelerometer data are the input of the function.

The Allan standard deviation of the measured gravitational acceleration was calculated in order to evaluate the post correction. Fig. 5 shows the results measured with the vehicle's engine off. After vibration noise correction, the standard deviation can reach 0.5 mGal with $T = 20$ ms and an average time of 10 s. Fig. 6 shows the results measured with the vehicle's engine on for generating electricity only. As the fringe fitting fails for $T = 20$ ms under this condition, we can only use $T = 10$ ms and $T = 5$ ms for gravity measurement with our gravimeter. In the case of $T = 10$ ms, the standard deviation is only 3 mGal with an average time of 100 s after correction, which is 2 times larger than that shown in Fig. 5(b), mainly due to the vibrations induced by the engine. Fig. 7

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TABLE III
THE SENSITIVITY AND SUPPRESSION FACTOR ACHIEVED IN VARIOUS EXPERIMENTAL CONDITIONS

Engine	Speed (km/h)	Half interrogation time T (ms)	Vibration induced phase noise $\sigma_{\Phi_{vib}}^{rms}$	Sensitivity before correction S_1 (mGal/Hz$^{1/2}$)	Sensitivity after correction S_2 (mGal/Hz$^{1/2}$)	Suppression Factor S_1/S_2
Off	0	20	1.36π	30.76	1.35	22.74
Off	0	10	1.35π	128.83	4.26	30.27
On	0	10	16.70π	1112.08	21.29	52.25
On	0	5	3.99π	1279.44	28.44	44.98
On	7	5	11.53π	3095.66	60.88	50.85

Fig. 8. The road (red dashed line) for the moving test.

illustrates the result measured when the vehicle was moving with a speed of 7 km/h. Only $T = 5$ ms can be used in this harsh environment. With respect to the experiments mentioned above, the fringe fitting results show that the delay τ_z is always around 2.2 ms. The contribution η_x and η_y, however, vary between -1.5% and 1.1%, which indicates the cross couplings due to the misalignment are very small.

The first 5 points of the curves shown in Fig. 5, 6 and 7 display a classical white noise behavior, which is characterized by a linear decrease with a slope of $-1/2$ in the log-log plot. Therefore, the sensitivity of the gravimeter under different conditions can be calculated by linear regression. The results are listed in Table III, and shown with the dash lines in Fig. 5, 6 and 7. We can use the suppression factor S_1/S_2 to evaluate the post correction, where S_1 is the sensitivity before correction, and S_2 is the sensitivity after correction. The standard deviations of the vibration induced phase noise $\sigma_{\Phi_{vib}}^{rms}$ are also listed in Table III, which agrees with the data shown in Table II. In the moving test, a suppression factor of 50.85 can be achieved, which improves the sensitivity to 60.88 mGal/Hz$^{1/2}$. With an averaging time of 100 s, the precision of the gravimeter may reach 6 mGal after the vibration noise correction.

It needs to be noted that there are other environmental factors that may affect the gravity measurement, which can make the above analysis inaccurate if their influence is significant. In static condition, we mainly need to consider the influence of the earth tide, which can induce a gravitational acceleration change of over 0.1 mGal in 24 hours. However, as our typical measurement time is only several hundred seconds, its influence is in μGal level, which is too small to influence the calculation results of the Allan standard deviation.

In the moving test, the Eötvos effect and the gravity fluctuation across the road should be taken into account. In this case, our vehicle moved along a straight road, which is shown with the red dashed line in Fig. 8 and has a length of 500 m

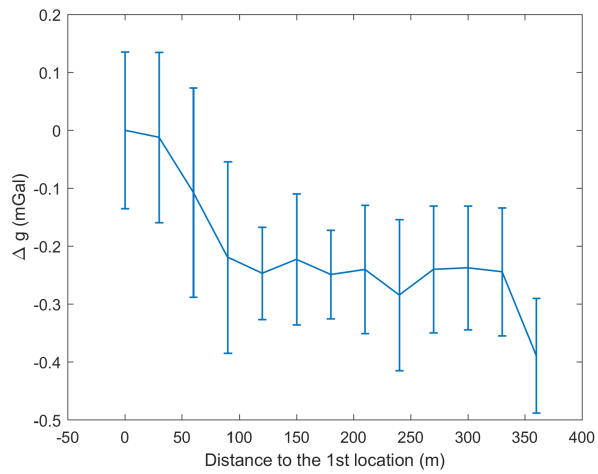

Fig. 9. The gravity fluctuation along the road relative to the gravity measured in the southernmost location (1st location). The length of each error bar represents two times the standard deviation of the data measured in the corresponding location.

approximately. With a speed of 7 km/h, the contribution of the Eötvos effect is estimated to be 20 mGal approximately using the equation described by Harlan [32]. But as the speed of the vehicle was nearly kept constant with a standard deviation of about 0.1 km/h (calculated from the GPS data), the fluctuation induced by the Eötvos effect is below 1 mGal, which is neglectable.

And to know the contribution of the gravity fluctuation across the road, we measured it with a commercial marine gravimeter (AT1M from Dynamic Gravity Systems). 13 locations were selected along the middle part of the road to place the spring gravimeter for the measurement. The distance between two adjacent locations is 30 m. In each location, the gravitational acceleration was measured for about 10 min. And after the measurement, we calculated the mean gravitational acceleration and the corresponding standard deviation for each location from the obtained data after tidal correction and tilt correction, which are illustrated in Fig. 9 with the solid line and error bars. Among the 13 locations, the southernmost location is defined as the first location. As we are only interested in the gravity fluctuation instead of the absolute gravitational acceleration, the mean value obtained in the first location has been deducted from all data shown in Fig. 9. The result indicates that the gravity fluctuation along the middle part of the road is less than 0.5 mGal, which is also negligible.

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Therefore, in the tests of our atom gravimeter, the influence of the environmental factors is not significant.

IV. Discussion

The sensitivity of an atom gravimeter scales quadratically with the interrogation time $2T$ [19]. Therefore, it's better to make it as large as possible to improve the precision of the gravity measurement. In static condition, previous tests show that the interrogation time of our atom gravimeter can reach 120 ms, which is mainly limited by the size of the vacuum apparatus. However, with the same sensor head, the interrogation time has been limited to 40 ms on the vehicle.

This is because we have to assure that the residual interferometric phase fluctuation is small enough to determine the gravitational acceleration after the vibration correction. When the vibration noise is measured with the accelerometer, the absolute error is dependent on the vibration noise level due to the performance of the accelerometer along with the associated electrical circuits. According to the comparison between our results shown in Fig. 4 and the PSD provided by Ménoret *et al.* [21], with the engine off, the vibration noise in our vehicle is much greater than that measured on the second floor of an inner-city building. And the environment is even worse in the moving test, which makes the vibration compensation challenging.

With a shorter interrogation time, the interferometric phase shift would be more sensitive to the vibration noise in the high frequency range, which is shown in Fig. 3. Though there are several air filled vibration mounts connected with the gyro-stabilized platform, which could somehow attenuate the vibration noise in the high frequency range, it may still require that the accelerometer has a high bandwidth to reduce the influence of the residual noise. The bandwidth (-3 dB) of our accelerometer is DC to 430 Hz, which is enough for vibration compensation in static condition according to the results given by Ménoret *et al.* [21]. However, it needs further investigation to estimate whether it is enough for the measurements on the vehicle. The normalized transfer function of the accelerometer would not be constant above several tens hertz, which would induce errors for measuring vibrations at higher frequencies.

Besides, in future research, the air filled vibration mounts should be changed to further suppress the vibration noise, mainly the noise whose frequencies are above several tens hertz. Due to the above reasons, this would be helpful to reduce the residual phase fluctuation after vibration compensation. The interrogation time may therefore be further increased, which is helpful to improve the precision of the gravity measurement.

The results shown here can be useful in many aspects. According to the acceleration spectrum and standard deviation given by Bidel *et al.* [8] and Geiger *et al.* [26], the vibration noise level in our vehicle is comparable with that in an aircraft. Therefore, the vibration compensation method may also be used for atom gravimeters designed for airborne or marine gravity surveys in future. And the results are also useful for developing other types of atomic sensors based on atom interferometry used in mobile environment, such as atomic gyroscopes [25] and accelerometers [26], [27], as they also use mirrors to implement the two Raman beams in a retro-reflected geometry.

V. Conclusion

A vibration compensation method for atom gravimeters has been summarized and demonstrated in this paper. Results show that this method can be used for our vehicle-mounted atom gravimeter, though the standard deviation of the vibration induced phase can be as large as 16.70π. The vertical vibration noise measured with the vehicle engine off is 3 times larger than that in the lab during daytime. In this static condition, we managed to achieve a sensitivity of 1.35 mGal/Hz$^{1/2}$ and a suppression factor of 22.74 with a half interrogation time of T = 20 ms after the vibration noise correction. And with an average time of 10 s, the standard deviation of the measured gravity data can reach 0.5 mGal. When the vehicle moves with a speed of 7 km/h, the vibration noise in the vehicle can be 2 orders of magnitude greater than that in the lab. In this condition, a suppression factor of 50.85 and a sensitivity of 60.88 mGal/Hz$^{1/2}$ were realized with T = 5 ms after the vibration compensation, which makes it possible to measure gravity continuously while the vehicle is moving. To our knowledge, this might be the first report about testing the atom gravimeter in a moving vehicle. And as there is a lack of detailed discussion about the application of the vibration compensation method in mobile environment, the results here would fill the gap of people's knowledge in this area and can be useful for developing high precision atom gravimeters for vehicle-borne, airborne or marine gravity surveys. And the compensation method can also be applied to other types of atomic sensors based on atom interferometry used in mobile environment, such as atomic gyroscopes and accelerometers.

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List of Abbreviation

DC	Direct current.
GPS	Global Positioning System.
IMU	Inertial measurement unit.
LPSD	Logarithmic frequency axis Power Spectral Density.
MEMS	Micro-electro mechanical system.
MOT	Magneto-optical trap.
PD	Photodiode.
PSD	Power spectral density.
UPS	Uninterruptible power supply

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